CLINICAL · EAR

Petrous apicitis

Petrous apicitis is infection that has tracked from the middle ear deep into the tip of the petrous bone. The classic picture is Gradenigo's triad: a discharging ear, deep pain behind the eye and a sixth nerve palsy. It is rare now, but it can appear just as the ear seems to be improving.

AT A GLANCE

- Spread of infection into a pneumatised apex.
- Gradenigo triad: discharge, deep eye pain, VI palsy.
- Needs CT and MRI together.
- Long IV antibiotics first; surgery for abscess or failure.
- Cholesterol granuloma is bright on T1 and T2; cholesteatoma restricts diffusion.

1 CHAPTER ONE Overview

Infection at the tip of the temporal bone.

1.1 Definition

Inflammation, and sometimes abscess, of the air cells of the petrous apex, spreading from acute or chronic otitis media and mastoiditis.

1.2 Why only some people get it

Only about 30% of petrous apices are pneumatised; the rest are diploic (marrow-filled) or sclerotic. Infection can only track into an aerated apex.

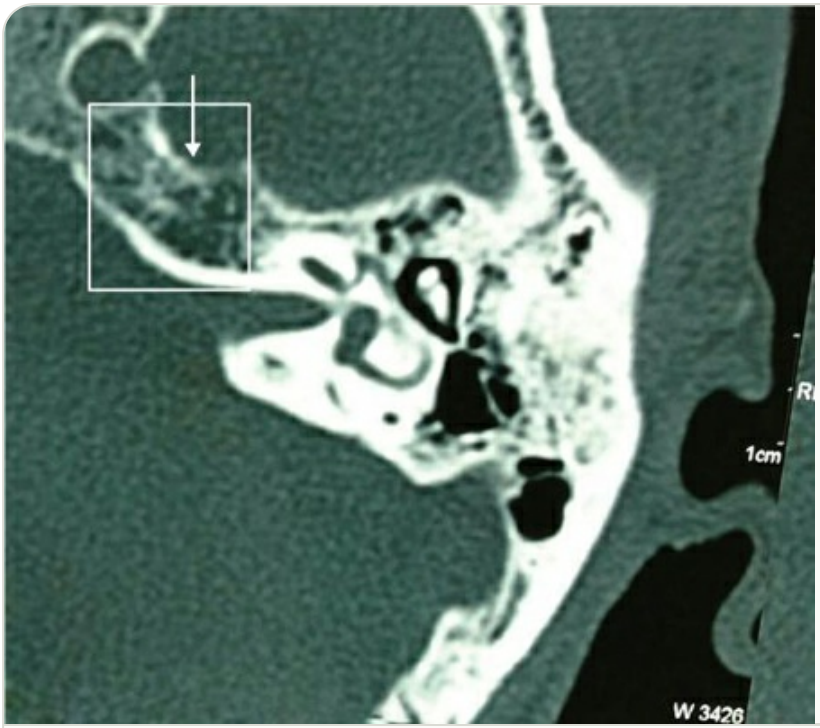


FIGURE 1 A pneumatised petrous apex.
Dhingra, fig. 1.15

1.3 Why it matters

The apex lies against the sixth nerve, the trigeminal ganglion, the internal carotid artery and the meninges. Infection there can cause meningitis, carotid injury and skull base osteomyelitis.

2 CHAPTER TWO Applied anatomy

Tracts into the apex.

2.1 Air cell tracts

The middle ear and mastoid connect with the apex by perilabyrinthine cell tracts:

- **Posterosuperior:** from the mastoid and antrum, above and behind the labyrinth, through *Trautmann's triangle* and the subarcuate region.
- **Anteroinferior:** from the hypotympanum and the region around the Eustachian tube, below the cochlea.

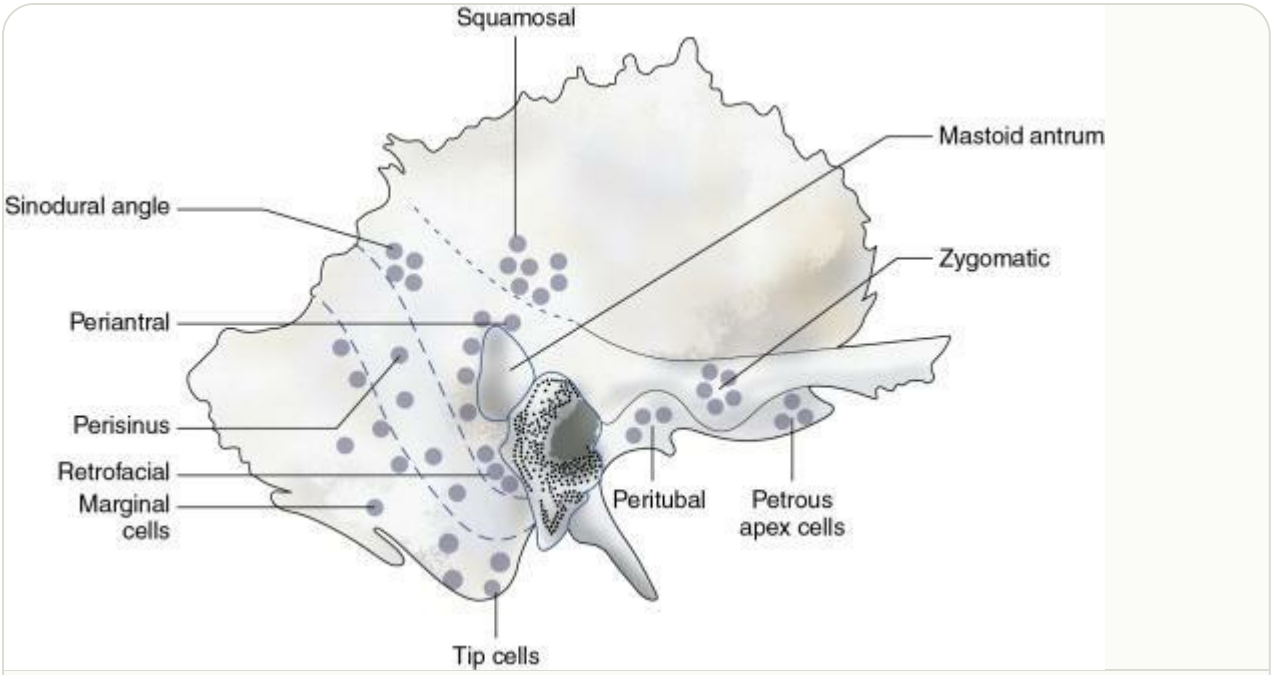


FIGURE 2 The air cell tracts of the temporal bone.
Dhingra, fig. 1.13

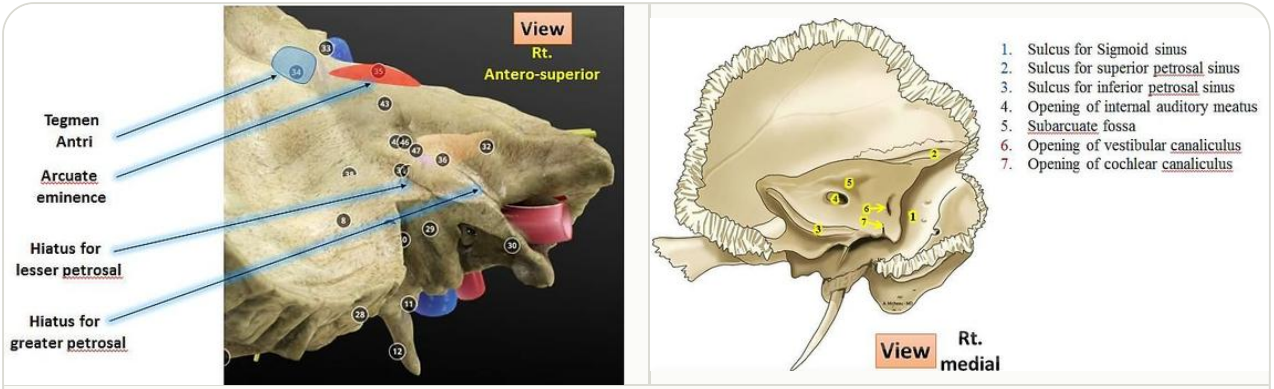


FIGURE 3 The anterior and posterior surfaces of the petrous bone.
Anatomy atlas · ENT Steps p. 21, ENT Steps p. 23

2.2 Neighbours

2.2.1 Dorello's canal

The sixth nerve passes under the petroclinoid (Gruber's) ligament, between the petrous tip and the sphenoid; inflammation here causes a lateral rectus palsy and double vision.

2.2.2 Meckel's cave

> The trigeminal ganglion sits in a depression on the apex; its irritation causes deep pain behind the eye.

3 CHAPTER THREE Clinical features

Gradenigo's triad.

3.1 The triad

- **Persistent ear discharge.**
- **Deep retro-orbital or facial pain,** from the trigeminal ganglion.
- **Diplopia from a sixth nerve palsy.**

The full triad is uncommon today; often only one or two features are present.

3.2 Other features

Fever, headache, neck stiffness and vomiting suggest meningeal spread; the seventh and eighth nerves are occasionally involved.

WHEN TO SUSPECT IT

Persistent discharge or deep pain despite an adequate mastoidectomy, or new pain after the ear has apparently settled.

4 CHAPTER FOUR Investigations

You need both CT and MRI.

4.1 High-resolution CT

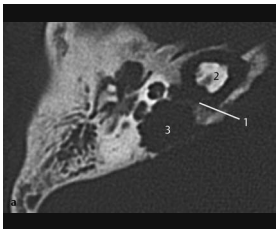
Shows the pneumatisation of the apex, breakdown of the cell septa and erosion of the bone.

4.2 MRI with contrast

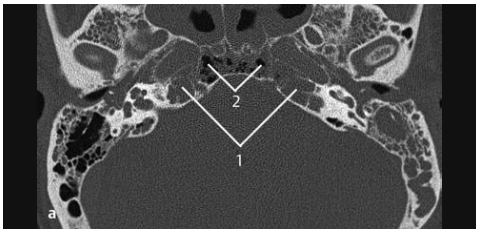
Separates marrow from fluid or pus, shows dural enhancement and intracranial spread, and assesses the carotid and cavernous sinus.

IMAGING IN THE APP 3 cases

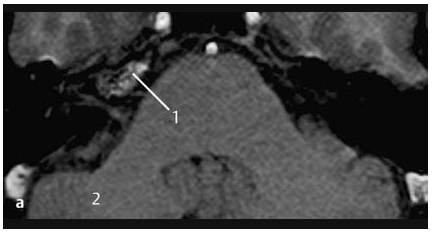
Petrositis, Gradenigo syndrome and apical pseudotumours.



Petrositis
Infection & inflammation · 3 images



Infection of the Skull Base and Gradenigo Syndrome
Infection & inflammation · 3 images



Pseudotumors in the Apex of the Petrous Bone
Tumours · 12 images

4.3 Differential diagnosis

- **Apical effusion:** fluid in an aerated apex, usually asymptomatic.
- **Asymmetric marrow:** a non-pneumatised apex full of fat that looks abnormal on MRI; a common false alarm.
- **Cholesterol granuloma:** bright on both T1 and T2, expanding slowly.
- **Cholesteatoma** of the apex.
- **Nasopharyngeal carcinoma** and other skull base tumours.

PETROUS APEX LESIONS ON MRI (OXFORD TEXTBOOK)

LESION	T1	T2	DIFFUSION AND ENHANCEMENT	CT
Asymmetric marrow (fat)	Bright, suppresses with fat saturation	Intermediate	No restriction	Normal bone, non-pneumatised apex
Trapped fluid or effusion	Dark	Bright	No restriction, no enhancement	Intact septa

LESION	T1	T2	DIFFUSION AND ENHANCEMENT	CT
Cholesterol granuloma	Bright	Bright	No restriction; rim only	Smooth expansion
Cholesteatoma or epidermoid	Dark	Bright	Restricted diffusion	Smooth erosion
Apicitis	Dark	Bright	Enhancing, with dural reaction	Destroyed septa
Chordoma or chondrosarcoma	Intermediate	Very bright	Heterogeneous enhancement	Bone destruction; clivus or petroclival fissure

Cholesterol granuloma is the commonest primary apex lesion, about ten times commoner than apical cholesteatoma, and forms when bleeding fills an obstructed air cell. Asymptomatic lesions are watched with serial imaging; symptomatic ones are drained, preserving hearing where possible, through an infracochlear or infralabyrinthine route, or endoscopically through the sphenoid when the lesion abuts it. Lower cranial nerve palsies with an apex lesion raise concern for malignancy.

5 CHAPTER FIVE Management

Antibiotics first.

5.1 Medical treatment

Admission, prolonged intravenous antibiotics that cover the organisms of otitis media and penetrate bone, and treatment of the middle ear source with myringotomy or mastoidectomy. Most acute cases now settle on antibiotics alone.

5.2 Surgery

For an abscess, bone necrosis, or failure of medical treatment. The route follows the cell tracts and the state of the hearing:

- **Hearing preserved:** through the mastoid via Trautmann's triangle or the subarcuate cells, infralabyrinthine or infracochlear routes, or through the middle cranial fossa.
- **Dead ear:** translabyrinthine or transcochlear routes, which give the widest access.

OPERATIVE GUIDES IN THEATRE 2 operations

Cholesterol Granulomas and Congenital Epidermoid Tumors of the Temporal Bone
Petrous apex lesions

Mastoid Surgery
Mastoid approach

6 CHAPTER SIX Complications

Neighbours at risk.

- Meningitis and other intracranial infection.
- Injury to or thrombosis of the internal carotid artery.
- Skull base osteomyelitis.
- Cavernous sinus thrombosis.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Gradenigo's triad,** and the anatomy behind each part: Meckel's cave and Dorello's canal.
- **Pneumatisation** of the apex in about 30%.

- **CT and MRI together.**
- **Surgical routes,** chosen by the hearing.

7.2 Common pitfalls

COMMON PITFALLS

- Reading asymmetric apical marrow as infection.
- Missing the diagnosis because the ear seemed to be getting better.
- Waiting for the full triad.

Written for this app from Oxford Textbook of Otolaryngology ch. 18, Oxford Specialist Handbook ch. 19, Dhingra ch. 1, 12 and K. J. Lee ch. 13, 22. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

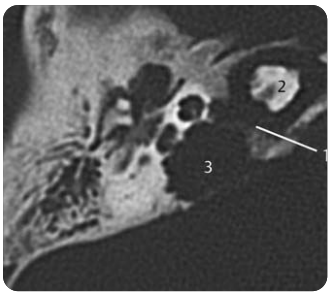
On imaging

The imaging of this topic: key findings, differentials and representative scans.

TEMPORAL BONE & SKULL BASE · INFECTION & INFLAMMATION

Petrositis

- Clinical symptoms may help differentiate between inflammatory disorders, asymptomatic pseudotumors, and other pathology. CT and MRI are complementary. Knowing whether the origin is intradural or extradural and the presence of bony changes help to establish the most likely diagnosis.

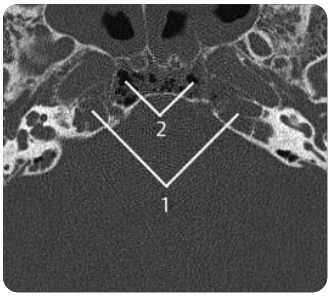


CT, axial. The internal auditory canal is seen (1), but is not clearly recognizable. Anteriorly, a rounded lucency in the petrous apex is visible…

TEMPORAL BONE & SKULL BASE · INFECTION & INFLAMMATION

Infection of the Skull Base and Gradenigo Syndrome

- In Gradenigo syndrome, the dural branches of the first division of the trigeminal nerve are involved in the symptoms of pain; the abducens nerve is paralyzed because of compression in the Dorello canal in the petrous apex. Pseudo-Gradenigo may be due to a nasopharyngeal carcinoma infiltrating the skull base.
- Malignant otitis externa may lead to skull-base infection. Immunocompromised and diabetic patients are prone to bacterial (e.g., Pseudomonas) or fungal (e.g.,) infections. Sequential radiographic and radionuclide (e. Aspergillus g., technetium and gallium) imaging may be used to monitor the efficacy of treatment. Complications may result from the spread of infection to adjacent intracranial or head and neck structures.



CT, axial. The petrous apex on both sides (1) is opacified. This could be because of normal fatty marrow but, considering the opacification in the…

TEMPORAL BONE & SKULL BASE · TUMOURS

Pseudotumors in the Apex of the Petrous Bone

IMAGING FINDINGS

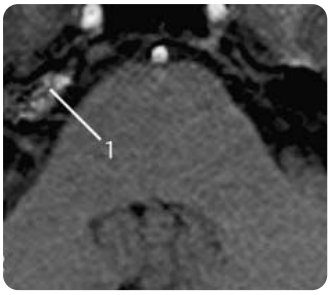
- Asymmetric marrow: bright on T1, suppresses with fat saturation, not expanded
- Trapped fluid: bright on T2, intact septa
- Cholesterol granuloma: bright on T1 and T2, expansile

TOP DIFFERENTIALS

- Cholesteatoma or epidermoid (DWI bright)
- Petrous apicitis (destruction, enhancement)
- Carotid aneurysm

CHECKLIST

- Compare with the other apex
- CT for expansion and cortex



MRI, T1-weighted, axial. T1-weighted image shows that the lesion (1) has high signal intensity compared with brain tissue (2). The contralateral…



Axial bone CT reveals an expansile cholesterol granuloma in the left PA with marginal bone dehiscence present. The right PA is well…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1152.